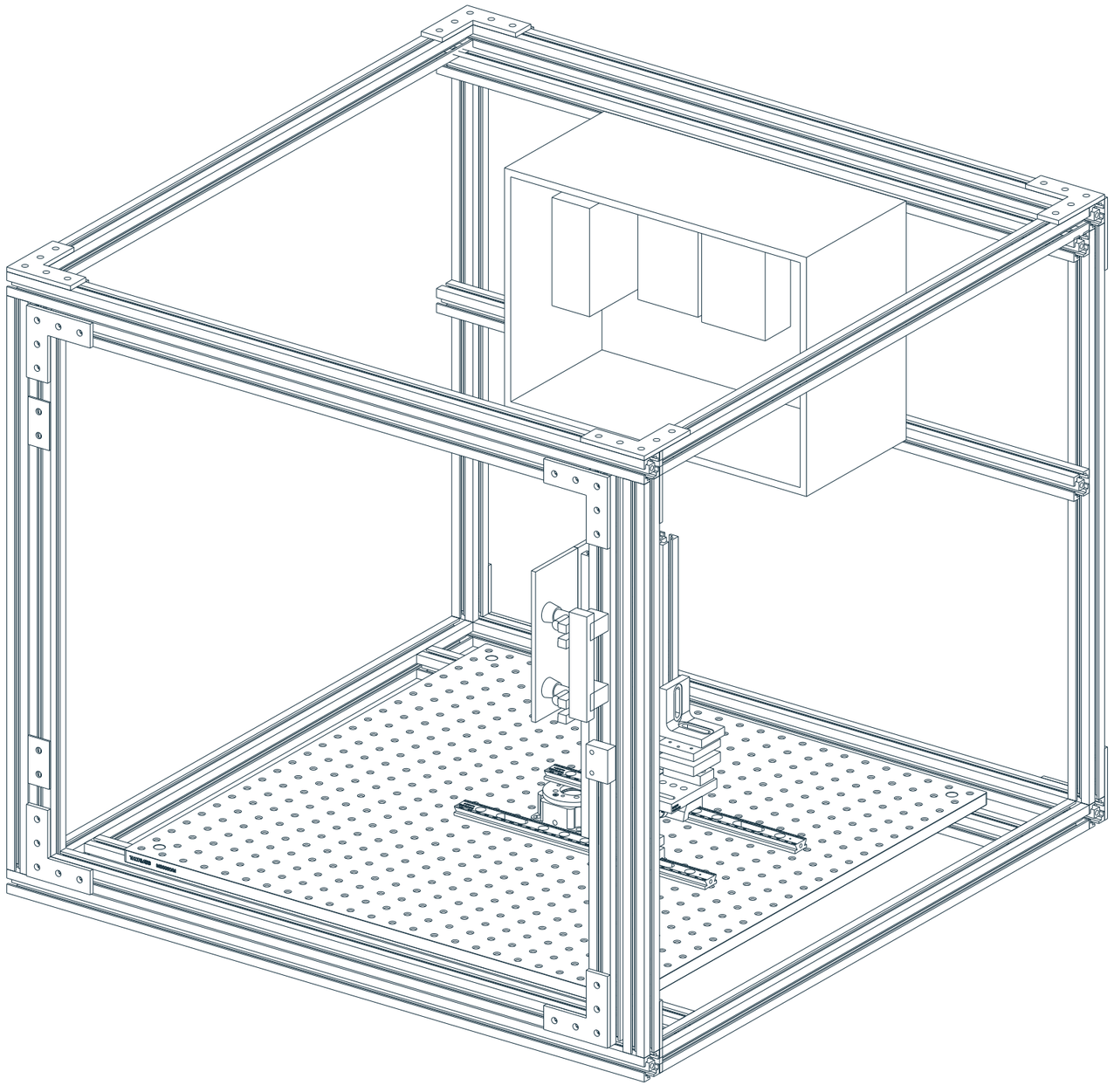
General arrangement

83 modeled components | Original vendor solids and identified reconstructed envelopes



DESIGN BASIS

762 x 762 x 660.4 frame; 25.4 mm 1010 profiles. Purchased optical breadboard and rotary stage. Passive purchased XY bearing. Four outside-surface vacuum contacts allow a natural closed pose.

DATUMS

Origin at frame footprint center. Z=0 is support surface; front is negative Y. Rotary axis at (12.5, 12.5, 38.1). Fixture-local coordinates are relative to that axis. Views are not print-scale templates.

MODEL LIMITS

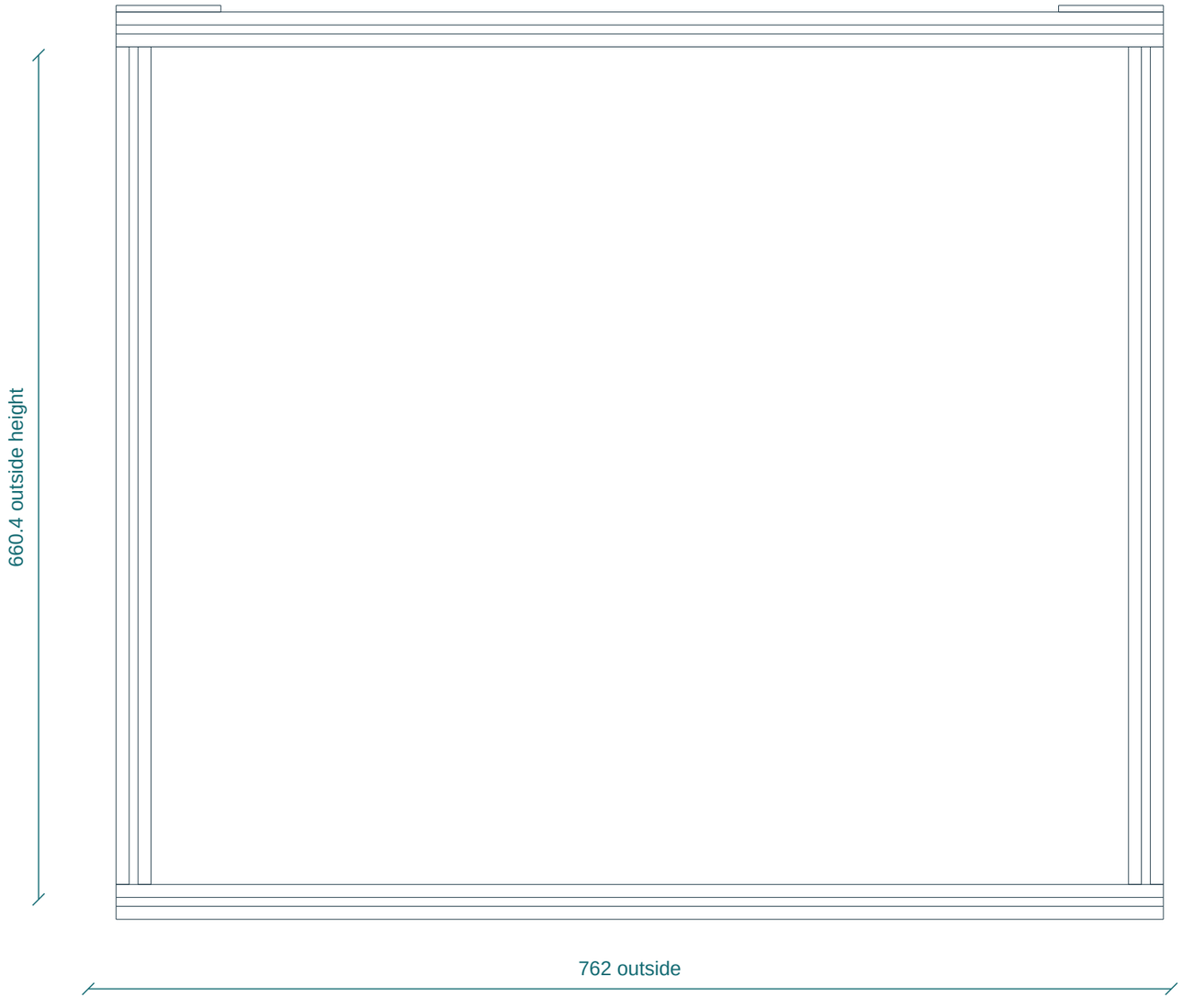
Fasteners, detailed cabinet mounts, latch/stop, pneumatic mounts and hose routes are not complete. A model count is not a complete BOM. Do not infer unmodeled mounting holes or a build release.

DOCUMENTS

This drawing book is separate from the 18-operation assembly manual. Electrical, pneumatic and fastener schedules provide the connection detail. The release register identifies every open hold.

Structure / purchased cut schedule

No shop machining, drilling, tapping or panel notching



Qty	1010 supplier cut	Use
4	762	Upper/lower long frame rails
6	711.2	4 perimeter + 2 board supports
4	609.6	Vertical posts
2	606.425	Door stiles
2	643.7252	Door crossbars
2	762	Cabinet supports; mount detail open
1	200	Moving grip mast
1	300	Fixed grip mast

Assembly targets: breadboard supports at Y=+/-275; top faces Z=25.4. Board top Z=38.1. Posts centered at X/Y=+/-368.3. Confirm supplier tolerances before accepting the decimal cut lengths; extra decimals describe nominal geometry, not guaranteed manufacturing precision.

Door / guard interface

Front view | catalog hinge and switch mounting dimensions

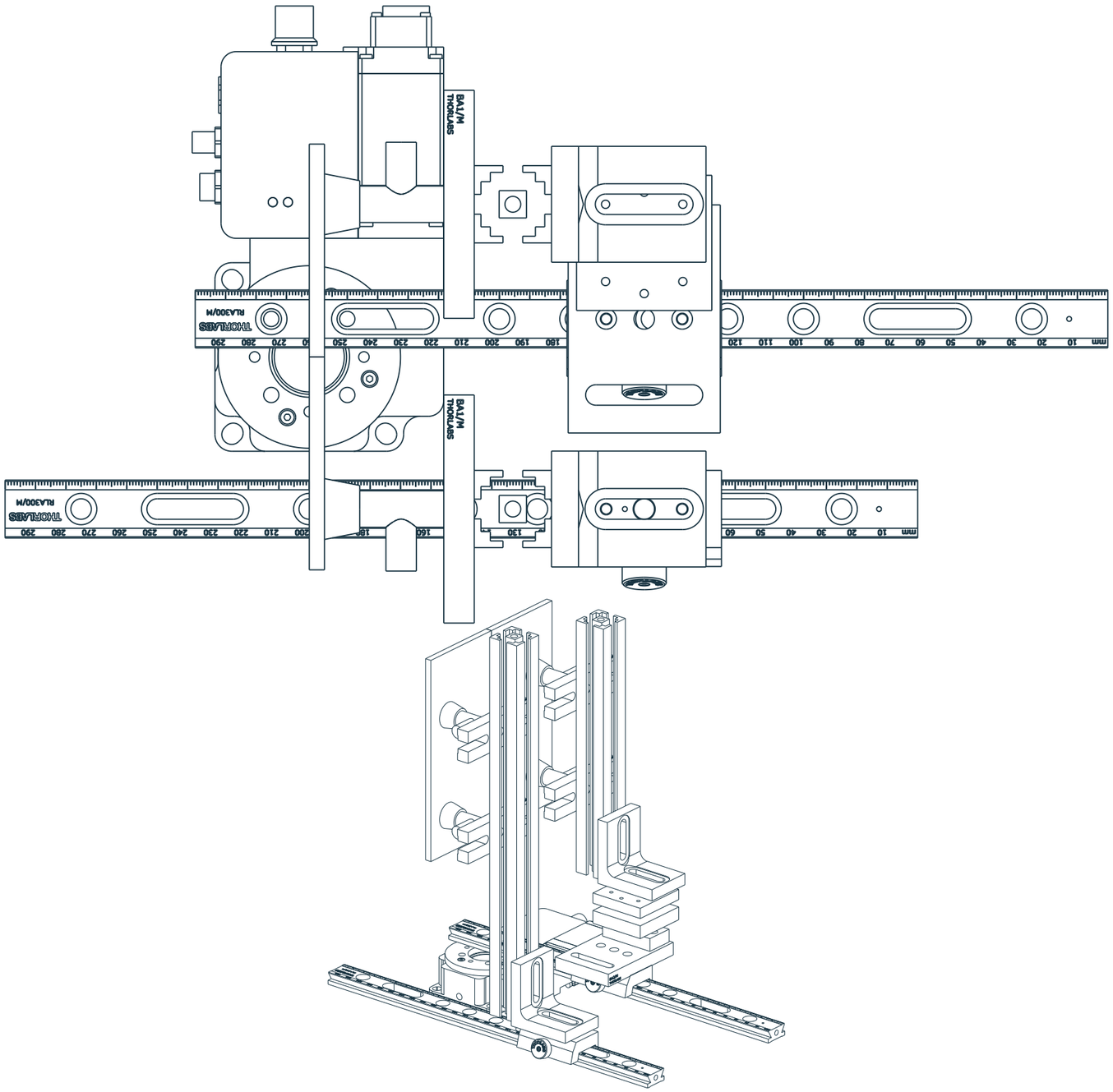


Feature	Nominal value / position
Door outside height	606.425
Left / right edge X	-354.0252 / +340.5
Bottom / top edge Z	26.9875 / 633.4125
Hinge center Z	150 and 510
Handle center Z / pitch	440 / 75.006
Switch mounting lines X	368.3 fixed / 327.8 door
Switch hole pitch / gap	22 / 2.5
Right opening	15.1; reach analysis required
Panel edge capture	6 nominal; gasket fit open

Front frame plates go on the inside/back face. Do not use switch housings as stops. The current 300 mm radial motion allowance leaves 193.0 mm to the nearest latch opening in plan; hoses, reach and stopping time still require verification.

Fixture / open pose and datums

Illustrative phone only | purchased rail, carrier and XY stack

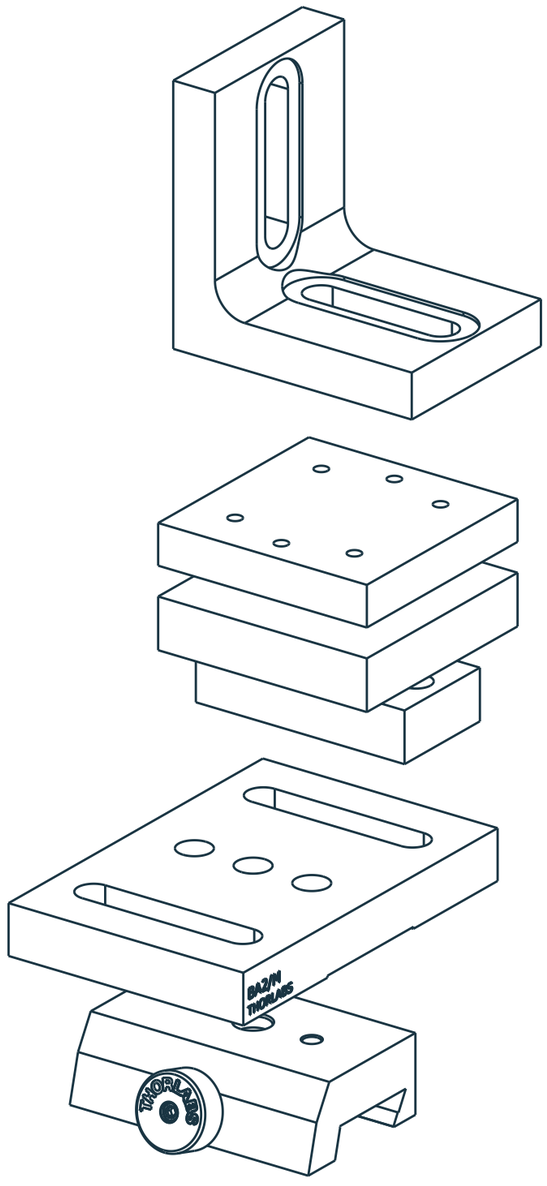


Fixture-local feature	Nominal mm
Rotating rail center X/Y	112.5 / 12.5
Rotating rail rear / front X	-37.5 / +262.5
Fixed rail center X/Y	50 / -50
Moving carrier center X	110.099
XY base center X/Y	110.099 / 37.5
XY total travel per axis	25.4; qualify middle 20
Cup contact plane X	5, surrogate only
Cup heights Z	160 / 240, surrogate only
Surrogate leaf dimensions	70 x 160 x 5 each

The actual phone is measured by the shop. The fixed and moving RC2 carriers are locked after setup; only the purchased plain XY stage floats. No hardware is placed between the closing display surfaces.

Moving interface / purchased mounting chain

Exploded view | separation is illustrative

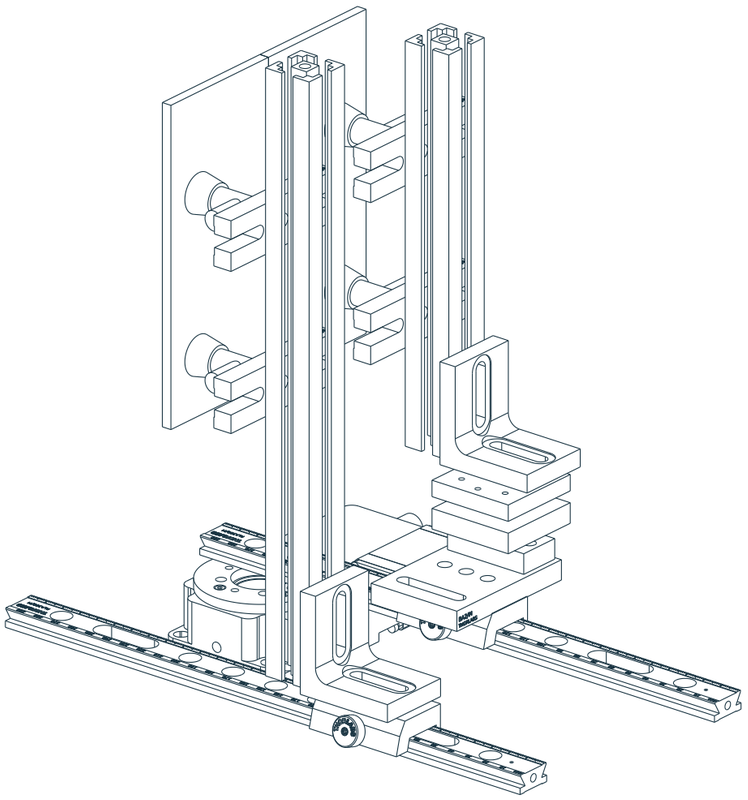


Order, bottom to top	Interface
RC2/M	Locked carrier; M4 threads
BA2/M + SD1	M4 x 14; 5.491 engagement
R451P-XY base	#6 through bolts; 33.3248 pitch
R451P-XY carriage	#6-32 imperial threads
AB90 + SD1	#6-32 x 5/8; bearing review
1010 mast	M6 T-slot connection
BA1/M / ESH / ESS	Exterior vacuum contacts

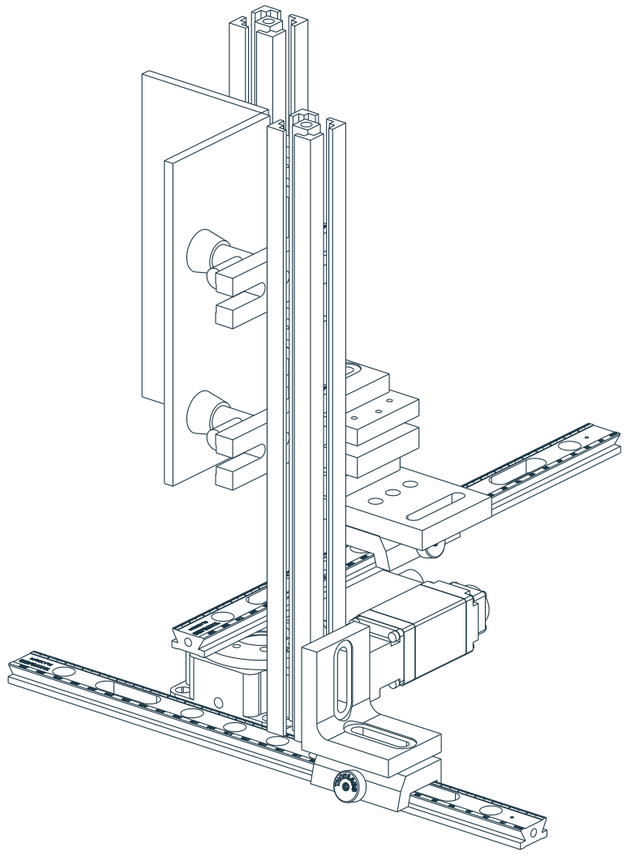
The BA2 slot nominally admits the two #6 shanks without machining. SD1 is marketed for #8/M4; the #6 head-bearing application remains M02. Do not treat this drawing as vendor approval of the mixed stack.

Natural closure / sampled motion study

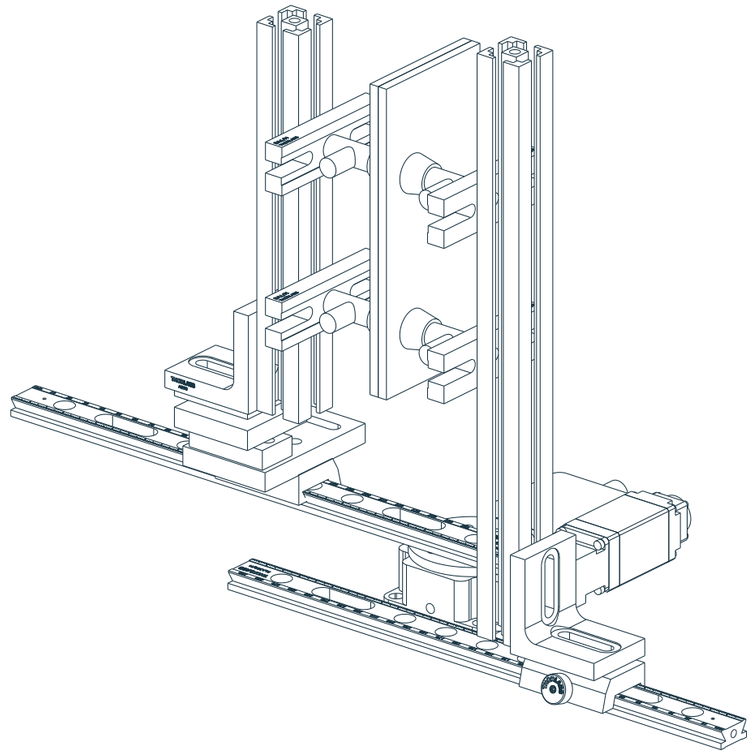
Motor rotation is not a direct measurement of phone hinge angle



OPEN / 0 degree stage rotation



MID / 90 degree stage rotation



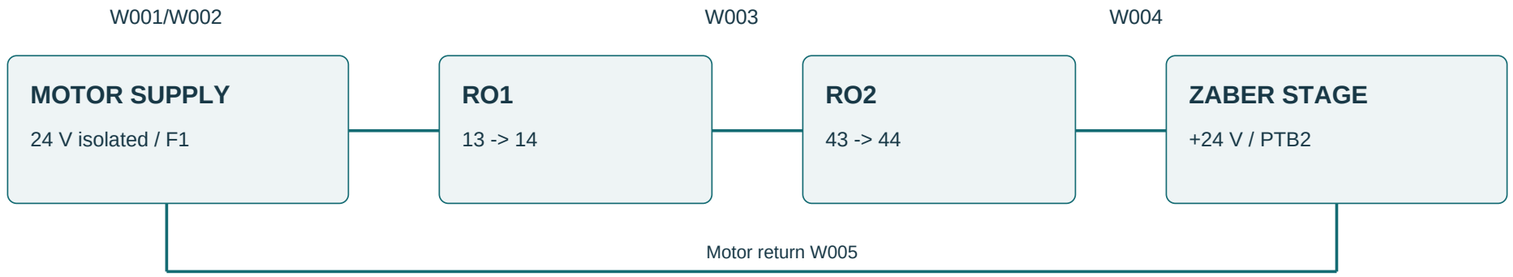
CLOSED / 180 degree stage rotation

Calculated evidence: the saved rigid-solid sweep checks 13 poses at 15 degree increments using B-rep intersection after bounding-box filtering. It found no overlap in those modeled moving/fixed solids. This is not continuous collision verification.

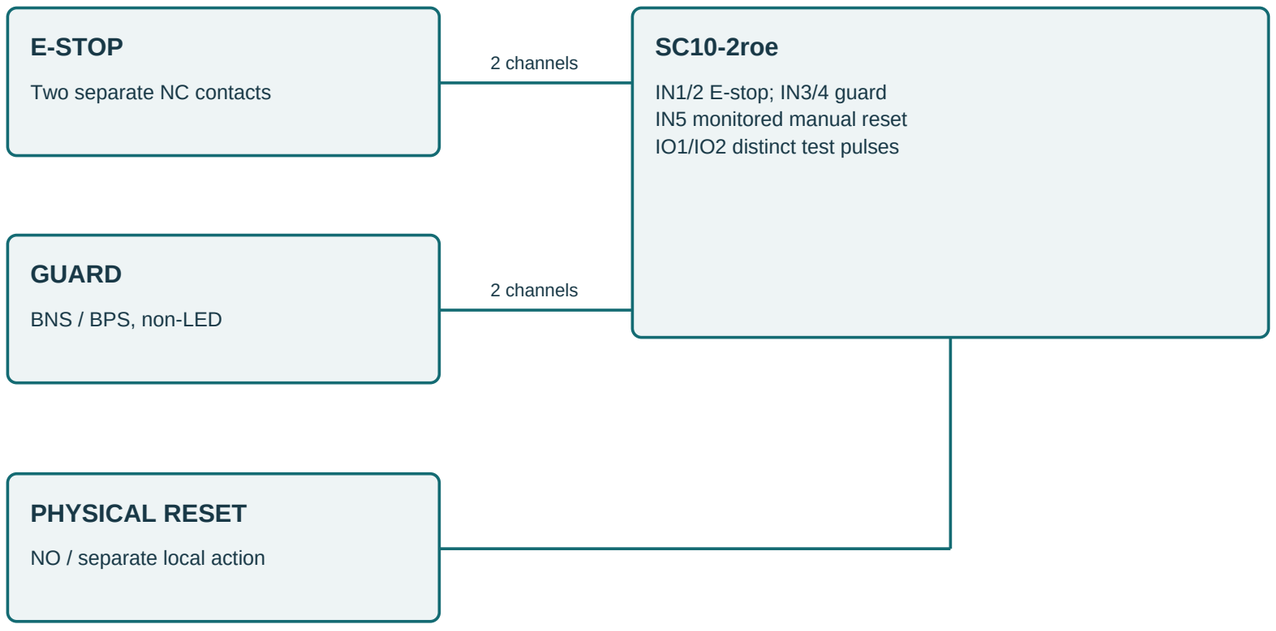
Excluded: actual phone compound hinge motion, XY displaced positions, fasteners, hoses, cables, between-sample clearance and stop dynamics. Full natural closure is an acceptance requirement; no overtravel is allowed to force an ideal drawing endpoint.

Electrical / motor disconnect architecture

Point-to-point details: electrical-installation.md | E01-E04 open



RO1 + RO2 enable only after both inputs healthy and reset



Vacuum remains applied on protective stop.

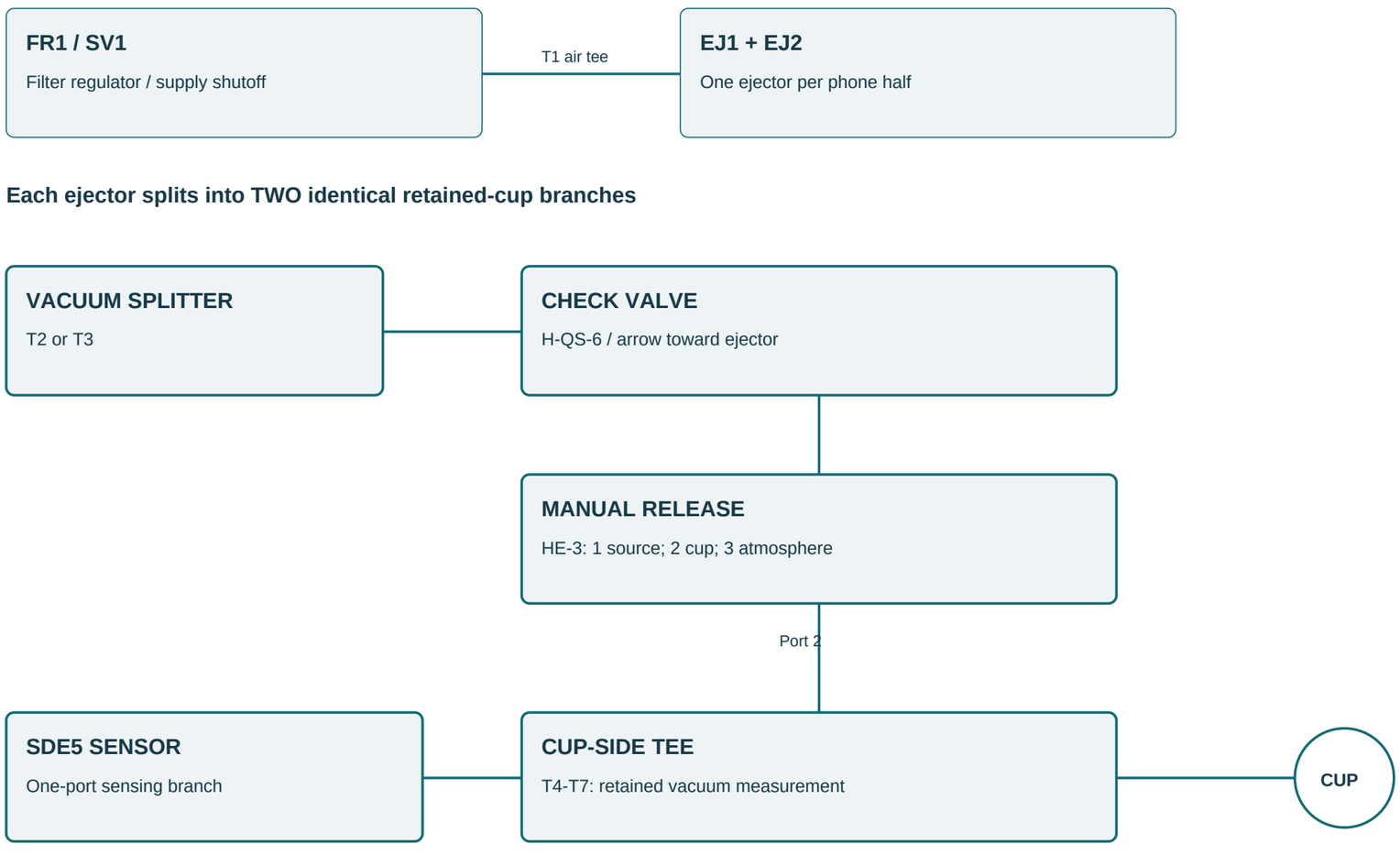
CONTROL POWER
Independent fused 24 V supply feeds SC10, digital I/O, temperature module and sensors. Pi uses its own purchased 5 V supply. All mains remains inside factory equipment.

PERMIT CONTACTS
Control +24 -> RO1:23-24 -> RO2:53-54 -> Waveshare DI1. DI COM -> 0 V. This is a process permissive; it does not turn the API into a safety controller.

RELEASE GATE
The actual Banner tool configuration and exported wiring report must verify input blocks, pulse-source support, terminal assignment, manual reset, no automatic restart and DC load rating.

Pneumatic / independently retained cup branches

Functional connection drawing | repeat retained branch four times



Repeat x4. Release only after both phone halves are supported.

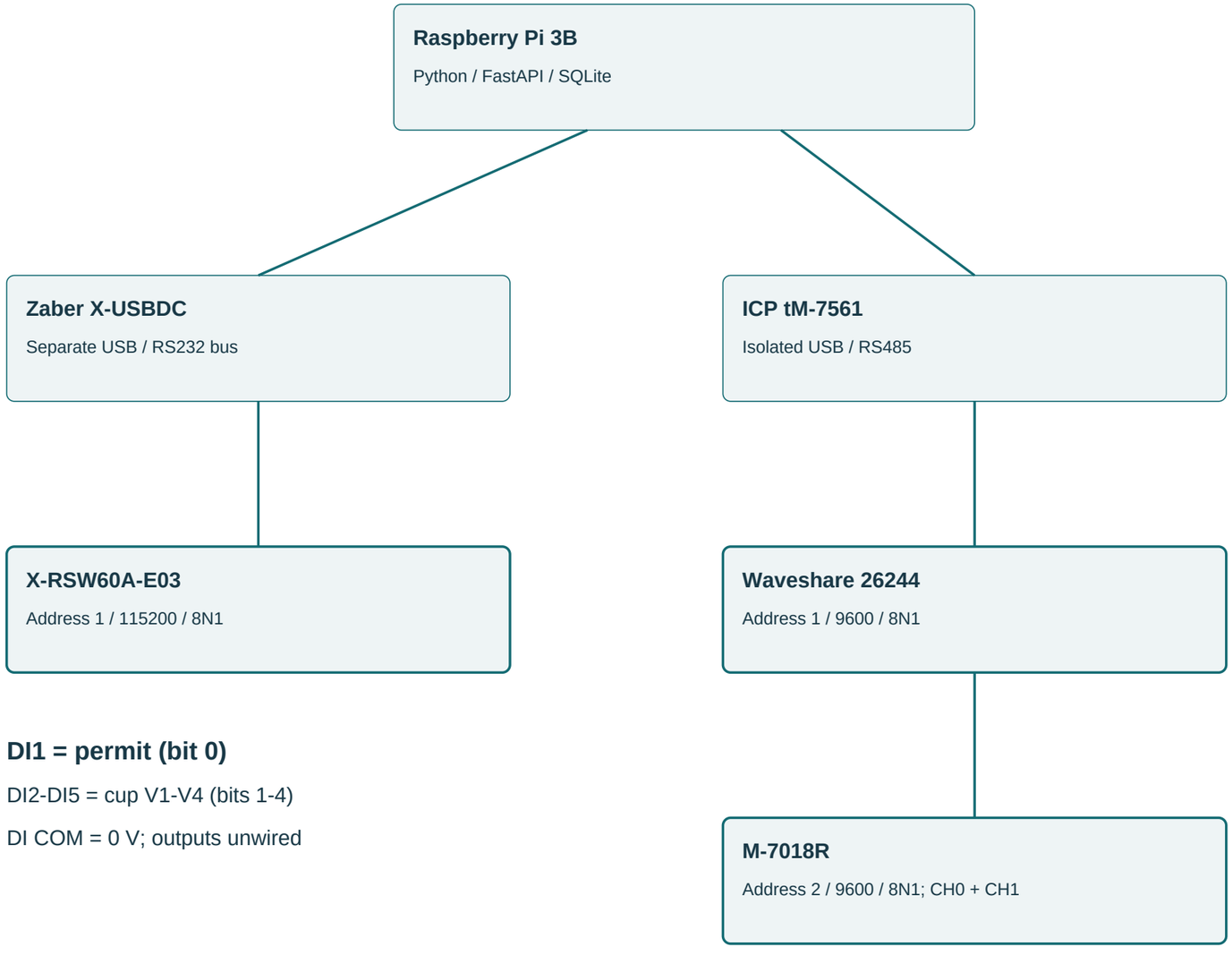
ORIENTATION
Check-valve airflow is cup -> ejector. SDE5 Q6E has one sensing port. Sense downstream of the release valve. The release valve vents only a physically supported phone.

ROUTING
6 mm OD tube; 50 mm minimum design bend radius for moving loops. Keep all tubing outside the display gap and qualified guard-access allowance. Final cuts come from the actual route.

QUALIFICATION
Setpoints and holding force cannot be taken from ideal cup area. Measure retention, peel, surface marking, leak response and supply-loss behavior on the actual contact surface.

I/O / process acquisition

Physical terminal labels and zero-based software bit map



Signal	Terminal	Bit/channel
Hardwired permit	DI1	0
Fixed lower vacuum	DI2	1
Fixed upper vacuum	DI3	2
Moving lower vacuum	DI4	3
Moving upper vacuum	DI5	4
Phone exterior Type K	CH0	0
Ambient Type K	CH1	1

DI1 = permit (bit 0)

DI2-DI5 = cup V1-V4 (bits 1-4)

DI COM = 0 V; outputs unwired

The software map uses zero-based wire addresses. Temperature engineering words are signed tenths of a degree C. Confirm firmware, channel type, CJC and open-wire behavior using actual hardware.

Fastener interface schedule

Installation proposal | unresolved rows remain holds; do not substitute nominal values for receiving checks

Joint	Qty	Purchased screw / associated part	Seat stack	Projection	Verification
4151 frame and door plates	80	80/20 3321 complete set	Vendor matched	Vendor matched	Five sets per plate; 16 plates
4119 brackets	24	80/20 3061 + 3382	Vendor matched	Vendor matched	Two per bracket; positions still under review
Door hinges	8	80/20 3386 complete assembly	Vendor matched	Vendor matched	Four per 2086 hinge
Door handle	2	80/20 3491	Vendor matched	Vendor matched	2062 handle
Rotary rail to stage top	2	ISO 4762 M6 x 8	2.540	5.460	M6 x 10 is too long; no substitution
Moving BA2/M to RC2/M	2	Bolt Depot 6395 M4 x 14 + SD1	8.509	5.491	RC2 usable thread depth 6 nominal
Fixed AB90 to RC2/M	2	Bolt Depot 6396 M4 x 16 + SD1	10.287	5.713	RC2 usable thread depth 6 nominal
Moving AB90 to XY top	2	Bolt Depot 15052 #6-32 x 5/8 + SD1	10.287	5.588	SD1/#6 head bearing requires approval M02

Controlling notes and manufacturer source links are in fastener-schedule.md. This schedule is part of the C-0.3 review package. Confirm polarity, thread form, actual seating and catalog compatibility before release. No custom adapter or machining is allowed.

Fastener interface schedule

Installation proposal | unresolved rows remain holds; do not substitute nominal values for receiving checks

Joint	Qty	Purchased screw / associated part	Seat stack	Projection	Verification
XY base to BA2 outer slot	2	Bolt Depot 15054 #6-32 x 7/8 + #6 nut/washer	Receiving check	Receiving check	Two through holes; do not tap either component
Holder to BA1/M	4	Bolt Depot 6417 M6 x 12	6.952	5.048	Measure usable blind thread; never bottom screw
BA1 outer slots to mast	8	M6 x 18 + DIN 125 M6 washer + 3873 nut	11.600	6.400	Confirm stock order code and nut engagement
AB90 vertical slot to mast	4	M6 x 16 + 3873 nut	8.255	7.745	Only 0.459 nominal slot-floor margin; review tolerance
Breadboard to support slots	4	M6 x 12 + 3873 nut	6.500	5.500	Approximately 3.29 usable engagement after slot lip; review
Stage base to breadboard	3	M6 x 16	5.500	10.500	Through-tapped 12.7 board; verify underside clearance
Fixed rail to breadboard	See rail drawing	M6 x 8 candidate	2.540	5.460	Final hole selection and thread strength review
BNS sensor and actuator	4	M4 x 20 + DIN 125 M4 washer + 3666 nut candidate	13.8	6.2	Verify molded seat, washer and slot engagement

Controlling notes and manufacturer source links are in fastener-schedule.md. This schedule is part of the C-0.3 review package. Confirm polarity, thread form, actual seating and catalog compatibility before release. No custom adapter or machining is allowed.

Electrical point-to-point schedule

Installation proposal | unresolved rows remain holds; do not substitute nominal values for receiving checks

Wire	From	To	Function
W001	Motor supply +	F1 input	Unswitched motor power
W002	F1 output	SC10 RO1:13	First motor disconnect
W003	SC10 RO1:14	SC10 RO2:43	Series link
W004	SC10 RO2:44	Stage power +, PTB2	Switched motor power
W005	Motor supply -	Stage power -	Motor return
W010	Control supply +	F2 input	Control power
W011	F2 output	X1.01	+24 V distribution
W012	Control supply -	X1.11	0 V distribution
W013	X1.02 / X1.12	SC10 +24 V / 0 V	Controller supply
W014	X1.03 / X1.13	Waveshare V+ / GND	Digital I/O supply
W015	X1.04 / X1.14	M-7018R +Vs / GND	Temperature I/O supply
W016	X1.15	Waveshare DI COM	Active-high PNP input reference
W020	SC10 IO1 test pulse	X2.01	Channel A fanout
W021	X2.01	E-stop NC contact A input	Independent channel A
W022	E-stop NC contact A output	SC10 IN1 via X2.03	E-stop A return

Controlling notes and manufacturer source links are in electrical-installation.md. This schedule is part of the C-0.3 review package. Confirm polarity, thread form, actual seating and catalog compatibility before release. No custom adapter or machining is allowed.

Electrical point-to-point schedule

Installation proposal | unresolved rows remain holds; do not substitute nominal values for receiving checks

Wire	From	To	Function
W023	SC10 IO2 test pulse	X2.02	Channel B fanout
W024	X2.02	E-stop NC contact B input	Independent channel B
W025	E-stop NC contact B output	SC10 IN2 via X2.04	E-stop B return
W026	X2.01	BNS black (S11)	Guard A source
W027	BNS blue (S12)	SC10 IN3 via X2.05	Guard A return
W028	X2.02	BNS white (S21)	Guard B source
W029	BNS brown (S22)	SC10 IN4 via X2.06	Guard B return
W030	X1.05	Physical reset NO input	Local reset supply
W031	Physical reset NO output	SC10 IN5 via X2.07	Monitored manual reset
W032	X1.06	SC10 RO1:23	Auxiliary permit supply
W033	SC10 RO1:24	SC10 RO2:53	Both relay banks required
W034	SC10 RO2:54	Waveshare DI1 via X2.08	Software motion permit

Controlling notes and manufacturer source links are in electrical-installation.md. This schedule is part of the C-0.3 review package. Confirm polarity, thread form, actual seating and catalog compatibility before release. No custom adapter or machining is allowed.

Pneumatic hose schedule

Installation proposal | unresolved rows remain holds; do not substitute nominal values for receiving checks

Labels	Qty	Connection	Starting allowance each
A01	1	Facility interface to FR1	2.0 m; interface remains site-specific
A02	1	FR1 to SV1	0.25 m
A03	1	SV1 to T1	0.25 m
A04-A05	2	T1 to EJ1/EJ2 supply	0.5 m
V01-V02	2	EJ vacuum to T2/T3	0.25 m
V03-V06	4	Splitter to CV1-CV4	0.25 m
V07-V10	4	Check valve to RV1-RV4 port 1	0.20 m
V11-V14	4	RV port 2 to cup sensing tee	0.20 m
V15-V18	4	Sensing tee to SDE5	0.20 m
V19-V20	2	Fixed tee to holder	0.40 m
V21-V22	2	Moving tee to holder	1.20 m service loop

Controlling notes and manufacturer source links are in pneumatic-installation.md. This schedule is part of the C-0.3 review package. Confirm polarity, thread form, actual seating and catalog compatibility before release. No custom adapter or machining is allowed.

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C101	4 ea	80/20 / 1010	762 mm frame rails; supplier cut	Selected
C102	6 ea	80/20 / 1010	711.2 mm rails; includes 2 breadboard supports	Selected
C103	4 ea	80/20 / 1010	609.6 mm posts; supplier cut	Selected
C104	2 ea	80/20 / 1010	606.425 mm door stiles; supplier cut	Selected
C105	2 ea	80/20 / 1010	643.7252 mm door crossbars; supplier cut	Selected
C106	2 ea	80/20 / 1010	762 mm cabinet support rails; supplier cut	Mounting detail hold M04
C107	1 ea	80/20 / 1010	200 mm moving grip mast; supplier cut	Selected
C108	1 ea	80/20 / 1010	300 mm fixed grip mast; supplier cut	Selected
C110	16 ea	80/20 / 4151	12 frame plates plus 4 door plates	Selected
C111	80 set	80/20 / 3321	Five screw/nut sets per 4151	Selected; CAD omits screws

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C112	12 ea	80/20 / 4119	8 base/support plus 4 cabinet corner brackets	Placement hold M04
C113	24 pair	80/20 / 3061 + 3382	Two screw/nut pairs per 4119	Stack review M04
C114	16 ea	80/20 / 3873	M6 T-nuts: 4 board + 4 mast angles + 8 cup plates	Selected
C115	4 ea	80/20 / 3666	M4 T-nuts for guard pair	Selected
C116	2 ea	80/20 / 2086	Metal door hinge	Selected
C117	8 set	80/20 / 3386	Hinge screw/nut assemblies	Selected
C118	1 ea	80/20 / 2062	Door handle; 75.006 mm hole pitch	Selected
C119	2 set	80/20 / 3491	Handle fastener assemblies	Selected
C120	3 ea	80/20 / 2643	723.2 x 621.6 x nominal 3 mm panels	Gasket fit hold M03
C121	1 ea	80/20 / 2643	723.2 x 723.2 x nominal 3 mm roof	Gasket fit hold M03

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C122	1 ea	80/20 / 2643	655.7252 x 567.625 x nominal 3 mm door	Gasket fit hold M03
C123	50 ft	80/20 / 2116	Push-in gasket; allowance includes corners	Panel retention hold M03
C124	1 ea	80/20 / 2053	Catalog deadbolt latch candidate	NOT SELECTED; gap fit M03
C125	1 ea	80/20 / 4483	Catalog door stop candidate	NOT SELECTED; mounting M03
C130	1 ea	Thorlabs / MB6060/M	600 x 600 x 12.7 breadboard; M6 25 mm grid	Vendor STEP obtained
C131	1 ea	Zaber / X-RSW60A-E03	60 mm rotary stage; vertical rotation axis in model	Vendor STEP obtained
C132	2 ea	Thorlabs / RLA300/M	300 mm rail; existing M6 mounting features	Vendor STEP obtained
C133	2 ea	Thorlabs / RC2/M	Locked setup carriers; M4 top threads	Vendor STEP obtained
C134	1 ea	Thorlabs / BA2/M	Moving stage adapter; existing slots	Vendor STEP obtained
C135	4 ea	Thorlabs / BA1/M	Cup-holder mounting plates	Vendor STEP obtained

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C136	2 ea	Thorlabs / AB90	Fixed and moving right-angle supports	Vendor STEP obtained
C137	6 ea	Thorlabs / SD1	Counterbore adapters; order pack sufficient for 6	Two #6 applications hold M02
C138	1 ea	Del-Tron / R451P-XY	Imperial plain XY; no micrometers springs or brackets	Factory configuration; drawing reconstruction
C140	2 ea	Bolt Depot / 6395	M4 x 14 socket head; BA2 to moving RC2	Nominal stack checked
C141	2 ea	Bolt Depot / 6396	M4 x 16 socket head; fixed AB90 to RC2	Nominal stack checked
C142	2 ea	Bolt Depot / 15052	#6-32 x 5/8 socket head; moving AB90 to XY	Adapter bearing hold M02
C143	2 ea	Bolt Depot / 15054	#6-32 x 7/8 socket head; XY base through BA2	Nut/washer stack hold M02
C144	4 ea	Bolt Depot / 6417	M6 x 12 socket head; cup holder mount	Blind depth inspection M02
C145	1 lot	Catalog fasteners / See fastener schedule	Remaining M6 and guard fasteners	Exact order-code/quantity hold M02
C150	1 ea	Durham / HDWB-3660-95	60 x 36 inch welded commercial workbench	Anchoring hold M05

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C151	4 ea	Bessey / TG4.012	12 inch capacity; 4 inch throat clamp candidate	NOT SELECTED; vendor drawing M05
C201	1 ea	Banner / SC10-2roe / 806222	45 W x 115 H x 100 D safety controller	Configuration export hold E04
C202	1 ea	Schmersal / BNS 260-02Z-L / 101184385	Non-LED dual NC guard switch	Selected
C203	1 ea	Schmersal / BPS 260-1 / 101184395	Matching guard actuator	Selected
C204	1 ea	Schneider / XALK178FH29	Complete dual-NC E-stop station	Mounting hold M04
C205	1 ea	Schneider / XALD102	Complete green NO reset station	Mounting hold M04
C206	1 ea	Waveshare / 26244	Modbus RTU IO 8CH; 72 x 121 x 34	Selected
C207	1 ea	ICP DAS / M-7018R	TC input; 76 x 120 x 42	Firmware/readback qualification Q03
C208	1 5-pack	Omega / SA1-K	Use two self-adhesive Type K probes	Exact lead-length option at order
C209	1 ea	ICP DAS / tM-7561	Isolated USB to RS485	Bus detail hold E03

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C210	1 ea	Raspberry Pi / SC0022 / 3 Model B	Headless controller; 85.6 x 56.5 board	Selected
C211	1 ea	Phoenix Contact / 2202874	RPI-BC 107.6 DEV-KIT KMGY; DIN case	Mounting hold M04
C212	1 ea	Adafruit / 1995	5 V 2.5 A factory micro-USB Pi supply	Selected
C213	2 ea	Zaber / PS12S-24V30	24 V 3 A; separate motor and control supplies	Output connector hold E01
C214	1 ea	Zaber / PTB2	Stage power terminal connector	Verify mating E01
C215	1 ea	Zaber / X-USBDC	Factory USB to RS232 stage lead	Selected; not T-USBDC
C220	1 ea	Altech / 127-912 / PC3625-16-m	Factory metric knockout enclosure; 361 x 254 x 165	Factory boss/mount hold M04
C221	1 set	Altech / 194-001 / MT1	Four mounting tabs	Mounting hold M04
C222	1 set	Altech / 190-005 / HK2	Catalog hinge kit	Selected
C223	2 ea	Altech / 198-336 / TS35/336	Precut 336 mm DIN rails	Factory boss layout hold M04

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C224	38 ea	Phoenix Contact / 3209510 / PT 2,5	20 distribution + 18 signal terminals; provisional	Final terminal allocation E02
C225	2 ea	Phoenix Contact / 3030213 / FBS 10-5	Separate +24 V and 0 V bridges	Selected
C226	2 ea	Phoenix Contact / 3211861 / PT4-HESI (5X20)	Fuse holders; 6.2 mm wide; fuses separate	Fuse selection E02
C227	1 lot	Catalog electrical / See electrical schedule	End plates clamps fuses glands wire labels microSD and cables	Unfinished order codes E02/E03
C301	4 ea	Festo / 189294 / ESS-20-SN	20 mm flat NBR suction cup	Phone surface qualification Q01
C302	4 ea	Festo / 189209 / ESH-HB-4-QS	28 mm M6 holder; 6 mm tube port	Blind mounting depth M02
C303	2 ea	Festo / 193488 / VN-05-H-T3-PQ2-VQ2-RO1	Continuous ejector; 14 mm wide; 97.6 mm long	Mounting hold P02
C304	4 ea	Festo / 153463 / H-QS-6	Individual cup check valves	Selected
C305	5 ea	Festo / 153475 / HE-3-QS-6	Four cup releases plus one supply shutoff	Mounting hold P02
C306	7 ea	Festo / 153129 / QST-6	6 mm tee; air and vacuum distribution	Selected

Purchased parts / selection register

Candidate and held items are explicit; not a complete purchase release | live URLs in assembly-parts.tsv

ID	Qty	Vendor / part	Description	Status
C307	4 ea	Festo / 527461 / SDE5-V1-O-Q6E-P-M8	One-port PNP normally-open vacuum switch	Threshold qualification P03
C308	4 ea	Festo / 8078223 / NEBA-M8G3-U-2.5-N-LE3	Factory straight M8 3-core 2.5 m cable	Selected
C309	20 m	Festo / 197384 / PUN-H-6X1-BL	6 OD / 4 ID tubing procurement allowance	Final route lengths P02
C310	1 ea	Festo / 529152 / MS4-LFR-1/4-D7-CRM-AS	G1/4 regulator; 5 um filter; gauge; manual drain	Bracket interface P01
C311	1 ea	Festo / 526064 / MS4-WR	Regulator bracket	Bracket interface P01
C312	1 ea	Festo / 532187 / MS4-WRS	Separate bracket nut	Selected
C313	2 ea	Festo / 186097 / QS-G1/4-6	Regulator fittings; verify purchasing pack size	Selected

Release evidence / open items

Design completion and physical qualification are separate

Gate	Subject	Required evidence
M01	Complete fixture and bench interference	Include fasteners, XY displaced positions, hoses, door swing and receiving tolerances; current rigid sample sweep is limited.
M02	Fastener schedule	Close all screw/nut/washer order codes, counts, bearing areas, blind depths, engagement and access; especially SD1 with #6 and breadboard mounts.
M03	Guard retention and closure	Confirm 2116 gasket with selected panel thickness; complete stock stop/latch and their mounting; verify access distances and stopping time.
M04	Cabinet and operator mounting	Resolve factory bosses, DIN rails, tabs, brackets, Pi case, glands and reset/E-stop mounting; add all hardware to CAD and BOM.
M05	Workbench anchoring	Verify purchased clamp geometry/load and retainers against commercial workbench; qualify frame restraint without drilling.
E01	Power connector compatibility	Verify factory PSU removable terminal construction, polarity and PTB2 mating without custom adapter.

No signature, pass result or physical measurement has been invented. The complete current source and saved tests travel with this revision. Close design holds before issuing build instructions; close physical qualification before unattended testing.

Release evidence / open items

Design completion and physical qualification are separate

Gate	Subject	Required evidence
E02	Electrical purchasing and cabinet fit	Select fuses, terminal end plates, clamps, ferrules, glands, wire, microSD and cables with exact quantities/order codes; verify thermal/clearance layout.
E03	RS485 installation	Pin exact converter manual, terminal labels, termination and reference/shield arrangement; complete cable order codes.
E04	Safety controller configuration	Generate and review the actual Banner configuration/exported wiring report, pulse-source support and reset logic; verify DC load rating.
P01	Regulator mounting	Resolve MS4-WR catalog bracket to 1010 mounting stack and access.
P02	Pneumatic mounting and route	Complete stock ejector/valve/sensor mounts, tube retainers, dynamic hose paths and as-built cut lengths.

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Release evidence / open items

Design completion and physical qualification are separate

Gate	Subject	Required evidence
Q01	Actual phone	Shop measures phone; qualify cup locations, coating compatibility, hinge path, allowable loads and full natural closure.
Q02	Mechanical acceptance	Measure bearing travel/drag, combined loads, retention, guarded access and protective-stop behavior with sacrificial specimen.
Q03	Hardware acceptance	Verify device identities/firmware, temperatures/open-wire faults, process restart, sensor loss, E-stop, guard and communications faults.
P03	Vacuum acceptance	Measure thresholds/hysteresis, leak response, supply-loss retention and supported manual recovery for each cup.

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